

Outline2

2026 年 6 月 26 日

講読資料

A First Course in Causal Inference (Part II, Chapter 3–5)

Preface:

1. Differences and similarities between **1. Fisherian inference (FRT in 1)** and **2. Neymanian inference (Neyman's theorem in 2)**
 2. Two kinds of inferences under **CRE and MPE*** with/without **stratification (discrete covariates, SRE)** or **regression (continuous covariates)**
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1 Completely randomized experiment (CRE) and Fisher randomization test (FRT)

- Treatment assignment mechanism (pp. 19)
- CRE
- **FRT (Fisherian inference)**
 - Sharp null hypothesis
 - Example of a general FRT
 - Canonical choice of test statistics (means, variance, distribution under H_0)
 - * Difference-in-means (Z test)
 - * Studentized stat. (T test)
 - * Wilcon rank sum test (non-parametric test)
 - * Kolmogorov-Smirnov test. (empirical distribution-based, see note of Order Statistics)
- Lady tasting tea

2 Neymanian Repeated sampling in CRE

- Finite population quantities
 - Mean, variance of potential outcomes, and ITE (individual treatment effects)
- **Neyman's theorem (Neymanian inference)**

- Thm 4.1 (unbiasedness, variance of difference-in-means estimator)
 - * Proof
- Thm 4.2 (limiting distribution of difference-in-means estimator)
- Regression analysis of CRE

3 Stratified random experiment (SRE) and post-stratification in CRE (discrete covariates)

- SRE (randomized block design)
 - Stratified randomization (block randomization)
 - propensity score
- FRT (stratified version)
 - stratified difference-in-means
 - stratified Studentized test
 - stratified rank sum test
 - stratified Kolmogorov-Smirnov test
- Stratified version of Neyman's inference
 - point estimate
 - confidence interval
- SRE vs CRE (remarks)*
- Post-stratification of CRE
 - Post-stratification
 - example

4 Rerandomization and regression (general/continuous covariates)*

- Rerandomization
 - Rerandomization using Mahalanobis distance (ReM)
- Regression and ANCOVA
 - Covariate-adjusted FRT

5 Matched-pairs experiments (MPE)*

- MPE
- FRT (paired version)
 - paired t test
 - Wilcoxon sign-rank test
 - KS test

- sign test
- Paired version of Neyman’s inference
- Covariate-adjusted
 - FRT
 - Regression adjustment
- MPE vs CRE (remarks)
- General matched experiment
 - FRT
 - Within-starta effects
 - A general causal estimand

6 Unification of Fisherian and Neymanian inferences